

# R.A.I. - COMPLETE TECHNICAL KNOWLEDGE BASE & VIVA MASTER MANUAL

## Smart India Hackathon (SIH) Problem Statement: SIH1521 - Explainable AI for Heavy/High-Impact Rainfall Prediction

Version: 1.0.0-PROD-FROZEN | Baseline Commit: b70c17a | Team Distribution: 6 Members

### Problem Statement: SIH1521 - Explainable AI for Heavy/High-Impact Rainfall Prediction

Version: 1.0.0 (Master Pre-Deployment Freeze Baseline)

Workflow: Local Python + React Development & Git/GitHub Version Control (NO DOCKER)

Target Audience: All 6 Team Members (Clear, Accurate, Defensible)

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## TABLE OF CONTENTS

1. [Chapter 1: The Big Picture (What is R.A.I. & Why Does It Exist?)](#chapter-1-the-big-picture)
2. [Chapter 2: Core Philosophy (Observe -> Predict -> Explain -> Act)](#chapter-2-core-philosophy)
3. [Chapter 3: Complete Technology Stack (Verified Inventory)](#chapter-3-complete-technology-stack)
4. [Chapter 4: Project Structure & File Manifest](#chapter-4-project-structure)
5. [Chapter 5: Frontend Architecture (5 Core Pillars & Radar Map)](#chapter-5-frontend-architecture)
6. [Chapter 6: Backend Architecture (FastAPI & API Endpoints)](#chapter-6-backend-architecture)
7. [Chapter 7: Authentication, OTP & Security Architecture](#chapter-7-authentication-otp--security)
8. [Chapter 8: Machine Learning Pipeline (XGBoost & Data Splitting)](#chapter-8-machine-learning-pipeline)
9. [Chapter 9: Model Performance Metrics & Imbalance Analysis](#chapter-9-model-performance-metrics)
10. [Chapter 10: Operational Decision Threshold (Why tau = 0.015?)](#chapter-10-operational-decision-threshold)
11. [Chapter 11: Explainable AI & TreeSHAP Mathematics](#chapter-11-explainable-ai--treeshap)
12. [Chapter 12: Warning Engine & Farmer AI Knowledge Hub](#chapter-12-warning-engine--farmer-ai-hub)
13. [Chapter 13: Data Ingestion (Open-Meteo & NASA GPM IMERG)](#chapter-13-data-ingestion)
14. [Chapter 14: Database Architecture (Supabase Cloud PostgreSQL)](#chapter-14-database-architecture)
15. [Chapter 15: Running the Application Locally](#chapter-15-running-the-application-locally)
16. [Chapter 16: "Why Did We Choose This?" (Key Architectural Decisions)](#chapter-16-why-did-we-choose-this)
17. [Chapter 17: Rapid-Fire Viva Question Bank & Answers](#chapter-17-rapid-fire-viva-question-bank)
18. [Chapter 18: Top 20 Must-Know Questions for the Presentation](#chapter-18-top-20-must-know-questions)
19. [Chapter 19: 6-Member Team Role Distribution](#chapter-19-6-member-team-role-distribution)
20. [Chapter 20: 1-Page Master Cheat Sheet (Memorize This!)](#chapter-20-1-page-master-cheat-sheet)

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### # CHAPTER 1: THE BIG PICTURE

#### 1. What is R.A.I.?

R.A.I. stands for **Rainfall Artificial Intelligence**.

It is an operational meteorological intelligence platform that predicts **heavy rainfall 24 hours in advance** and explains **WHY** the model triggered that risk using **TreeSHAP Explainable AI**.

## 2. What problem does it solve?

- Extreme localized rainfall (cloudbursts, intense convective storms) causes urban waterlogging, flash floods, and agricultural devastation across India.
- Standard weather applications provide broad regional forecasts but fail to identify localized extreme precipitation events and treat predictions as an opaque black box.
- R.A.I. provides transparent, physics-grounded early warnings with exact feature attributions for citizens, emergency response authorities, and farmers.

## 3. What is the SIH Problem Statement?

- **Problem Statement Code:** `SIH1521`
- **Title:** `*"Explainable AI for Heavy/High-Impact Rainfall Prediction"`
- **Target Event Definition:** Heavy rainfall accumulation  $\geq 64.5$  mm/day (the official India Meteorological Department / IMD classification standard).

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### # CHAPTER 2: CORE PHILOSOPHY

```

1. OBSERVE (Layer 1)
Fetch high-resolution atmospheric physics from Open-Meteo & NASA GPM satellites.
|
v
2. PREDICT (Layer 2)
Run calibrated XGBoost ML model to compute heavy rainfall probability p(x).
|
v
3. EXPLAIN (Layer 3)
Apply TreeSHAP to compute exact additive feature attributions (positive & negative).
|
v
4. ACT (Layer 4)
Map risk score to IMD-aligned operational warning levels & actionable checklists.

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### # CHAPTER 3: COMPLETE TECHNOLOGY STACK

#### 1. Frontend Technologies (`src/`)

- **React (v18.3.1):** Component-based UI framework.
- **TypeScript (v5.6.3):** Strict static type checking across data contracts.
- **Vite (v6.0.1):** High-speed frontend build tool and dev server (`http://localhost:3000`).
- **React Router DOM (v6.28.0):** Client-side routing and protected route guards.
- **Leaflet (v1.9.4):** Interactive mapping engine for city markers and radar overlays.
- **Lucide React (v0.468.0):** Clean, lightweight iconography.
- **Vanilla CSS Modules:** Scoped component styles avoiding global collisions.
- **Custom Design Tokens:** Light futuristic design system (Apple / Linear style).

#### 2. Backend Technologies (`ml/`)

- **Python (3.14.6 / 3.11+):** Core language for data science and API services.
- **FastAPI (v0.141.1):** High-performance asynchronous REST API framework (`http://127.0.0.1:8000`).
- **Uvicorn (v0.52.3):** Production ASGI web server running FastAPI.
- **Pydantic (v2.13.4):** Request and response data contract validation.
- **Requests & Httpx:** HTTP client libraries for external meteorological ingestion.
- **Pytest (v9.1.1):** Automated test runner executing 26 unit and integration test cases.

#### 3. Machine Learning & Explainable AI

- **XGBoost (v3.4.1):** Gradient Boosted Decision Trees trained on 175,440 hourly records.
- **Scikit-Learn (v1.9.0):** Platt Sigmoid probability calibration and metric evaluation.

- **SHAP (TreeSHAP v0.52.0)**: Exact game-theoretic Shapley additive feature attribution engine.
- **NumPy & Pandas**: Tabular feature processing and rolling window calculations.
- **Joblib (v1.5.3)**: High-performance model serialization (`model.pkl`, `calibrated\_model.pkl`).

#### 4. Cloud Database, Auth & External APIs

- **Supabase Cloud PostgreSQL**: Managed cloud relational database (6 tables).
- **PostgREST API**: RESTful query interface for Supabase.
- **Brevo Transactional API**: Email gateway delivering 6-digit OTP codes via verified Sender ID `1` (`RAINAIWORK@GMAIL.COM`).
- **Open-Meteo REST API**: Hourly 2m temperature, dew point, surface pressure, cloud cover, wind, precipitation.
- **NASA GPM IMERG 0.1 deg Grid**: Spaceborne microwave/IR precipitation data layer.
- **RainViewer Doppler Radar**: Live animated Doppler radar tile layer for Leaflet.

#### 5. Collaboration & Version Control (NO DOCKER)

- **Git**: Local version control tracking all commits and baselines.
- **GitHub**: Remote repository for team collaboration, code reviews, and synchronization.

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#### # CHAPTER 4: PROJECT STRUCTURE

```
d:/rainainew/
|-- src/ # React 18 + TypeScript Frontend
| |-- pages/ # 5 Core Pillar Pages
| | |-- Home/ # Landing page with dual CTAs & telemetry preview
| | |-- RiskMap/ # MAIN SHOWCASE: 4-Layer Inspector & Doppler radar
| | |-- Dashboard/ # Central risk and location telemetry
| | |-- Emergency/ # Civil defense & river flood gauges
| | |-- Relief/ # Community shelters & logistics
| | |-- Farmer/ # Farmer command center & crop calendar
| | |-- Intelligence/ # [PROTECTED] Claude AI Assistant
| |-- components/ # UI components, layout shell, and map components
| |-- context/ # AuthContext, LocationContext, FarmerContext
| |-- data/raiKnowledge/ # 35,857 agronomic records across 18 JSON files
|-- ml/ # FastAPI + Machine Learning Backend
| |-- server.py # Main FastAPI server entry point
| |-- inference/service.py # Feature assembly and XGBoost inference coordinator
| |-- explainability/shap_engine.py # TreeSHAP calculation engine
| |-- features/registry.py # 27-feature formal meteorological registry
| |-- auth/otp_engine.py # Brevo OTP generation, salted HMAC, and verification
| |-- db/client.py # Supabase Cloud PostgreSQL client
| |-- models/rainfall_model_v1/ # model.pkl, calibrated_model.pkl, metrics.json
| |-- tests/ # 26 automated pytest unit & integration tests
|-- docs/ # SIH1521 technical documentation, PDF, and HTML
-- package.json # Frontend dependency manifest
```

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#### # CHAPTER 5: FRONTEND ARCHITECTURE (5 PILLARS)

##### 1. Pillar 01 - Dashboard (`/dashboard`):

- Central location telemetry displaying real-time temperature, humidity, pressure, and risk summary for the selected city.

##### 2. Pillar 02 - Interactive Risk Map (`/risk-map`) [MAIN SHOWCASE]:

- Real-time animated **RainViewer Doppler radar precipitation tiles**.
- **4-Layer Telemetry Inspector**:
  - **Layer 1: Observed**: Real physical measurements (Temperature, Dew Point, Humidity, Pressure, Wind).
  - **Layer 2: Predicted**: Calibrated probability  $p(x)$  against operational threshold  $\tau = 0.015$ .
  - **Layer 3: Explained**: TreeSHAP horizontal bar chart (factors increasing risk in orange/red, protective factors in blue).
  - **Layer 4: Operational**: Warning status (**NO\_WARNING**, **ADVISORY**, **WATCH**, **WARNING**) and action checklists.

##### 3. Pillar 03 - Emergency Civil Defense (`/emergency`):

- River flood gauge monitoring, danger water levels, and emergency helpline routing.

#### 4. Pillar 04 - Relief Logistics (/relief):

- Verified community relief shelters, dry storage zones, and logistics supply routes.

#### 5. Pillar 05 - Farmer Command Center (/farmer):

- Crop growth stage tracking, soil-grounded irrigation watch, and agronomic risk advisories.

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### # CHAPTER 6: BACKEND ARCHITECTURE & APIS

The FastAPI backend runs on <http://127.0.0.1:8000>.

#### Key Endpoints:

- ``GET /api/prediction/heavy-rainfall?latitude=...&longitude=...&city=...``: Executes live weather ingestion, feature extraction, calibrated inference, and TreeSHAP attribution.
- ``GET /api/model/status``: Returns model version (``v1.0.0-sih-xgb``), threshold (``0.015``), and training lineage.
- ``GET /api/model/explainability``: Returns global feature importance rankings across all 27 registered features.
- ``POST /api/auth/register/initiate``: Creates user challenge and dispatches 6-digit OTP via Brevo email.
- ``POST /api/auth/otp/verify``: Validates OTP hash, marks account verified, and issues session token.
- ``POST /api/auth/login``: Authenticates returning users with 15-minute brute-force lockout protection.

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### # CHAPTER 7: AUTHENTICATION, OTP & SECURITY

#### 1. Cryptographic Salted OTP:

- 6-digit random code hashed using **HMAC-SHA256** with server-side salt.
- Dispatched via Brevo API using verified Sender ID **1 (RAINAIWORK@GMAIL.COM)**.
- Strict 5-minute expiry, max 5 attempts, and immediate single-use destruction (`is_consumed = TRUE`).

#### 2. Password Security:

- Salted SHA-256 with server pepper. Zero passwords stored in browser storage.

#### 3. Session Management:

- 256-bit cryptographic hex tokens (`secrets.token_hex(32)`).
- Remember Me checked `\to localStorage` (email string only).
- Remember Me unchecked `\to sessionStorage` (destroyed on tab close).

#### 4. IDOR Tenant Defense:

- Supabase authorization gates strictly isolate citizen vs. farmer plot datasets.

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### # CHAPTER 8: MACHINE LEARNING PIPELINE

#### 1. Dataset & Zero-Leakage Partitions

The model was trained on **175,440 hourly meteorological records** spanning 2023 to 2024 across 10 diverse Indian climate zones:

- **Training Set (70%)**: 122,800 records (Jan 1, 2023 to May 26, 2024).
- **Validation Set (15%)**: 26,320 records (May 26, 2024 to Sep 13, 2024) - \*Used for threshold selection and probability calibration\*.
- **Test Set (15%)**: 26,320 records (Sep 13, 2024 to Dec 31, 2024) - \*Held out for final performance verification\*.

#### 2. Feature Inventory (14 Production vs. 27 Registered Features)

- **14 Production Inference Features**: The exact vector passed to ``model.pkl`` (surface measurements, humidity, pressure, wind, past rainfall sums, evapotranspiration, and temperature extremes).

- **27 Registered Feature Schema:** The complete formal meteorological schema in `ml/features/registry.py` including backward lags (1h, 3h, 6h, 12h, 24h), satellite accumulations, 3-hour physical tendencies, thermodynamic proxies (`dew\_point\_spread`, `convective\_energy\_proxy`), and solar/monsoon harmonics.

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# CHAPTER 9: MODEL PERFORMANCE METRICS

| Metric Name                          | Value      | Correct Viva Interpretation                                                                                                                                                            |
|--------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| **ROC-AUC**                          | **0.9605** | <b>**Discrimination Power**</b> : Indicates exceptional ranking capability between heavy rain events and non-heavy rain days across all thresholds. *(Do not call this 96% accuracy)*. |
| **PR-AUC**                           | **0.1082** | <b>**Precision-Recall Area**</b> : Given an extreme baseline incidence of 0.44%, this represents a <b>**24.6x lift over random guessing**</b> .                                        |
| **Calibrated Brier Score**           | **0.0044** | <b>**Probability Reliability**</b> : Mean squared error between predicted probability and actual outcome. Rated <b>**EXCELLENT**</b> .                                                 |
| **Validation Recall (tau=0.015)**    | **52.1%**  | <b>**Early Warning Catch Rate**</b> : Successfully captures 52.1% of extreme rainfall events 24 hours ahead.                                                                           |
| **Validation Precision (tau=0.015)** | **10.96%** | <b>**Alert Reliability**</b> : 1 in 9 alerts corresponds to >= 64.5 mm rain, with the remainder bringing moderate-to-heavy showers.                                                    |
| **Validation F2 Score**              | **0.2975** | Weights flood recall twice as heavily as false alarms.                                                                                                                                 |

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# CHAPTER 10: OPERATIONAL DECISION THRESHOLD (tau = 0.015)

- **The Class Imbalance Challenge:** Heavy rain (>= 64.5 mm/day) occurs on only **0.44% of days (1 in 227)**.
- **Why tau = 0.015 instead of 0.50?**: A default 50% cutoff predicts "NO RAIN" 365 days a year (0% recall). A calibrated probability of 1.5% represents a **3.4x surge over the historical base rate**.
- Setting tau = 0.0150 on validation data catches **52.1% of flood events** while triggering alerts on only 7.32% of days.

**Risk Boundaries:**

- **LOW** (p < 0.75\%): Atmospheric conditions unfavorable for heavy rain.
- **MODERATE** (0.75\% <= p < 1.5\%): Elevated moisture/instability; heightened monitoring.
- **HIGH** (1.5\% <= p < 5.0\%): **Threshold Breached [WATCH]** - High heavy rain probability.
- **CRITICAL** (p >= 5.0\%): **Severe Danger [WARNING]** - Potential cloudburst / flood.

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# CHAPTER 11: EXPLAINABLE AI & TreeSHAP

**1. What TreeSHAP Does:**

TreeSHAP computes the exact additive contribution of each atmospheric feature to the model's predicted risk score using game-theoretic Shapley values:

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i$$

- $\phi_0$  = -4.852 log-odds (the expected average baseline across India).
- $\phi_i > 0$ : Feature increased the model's predicted risk (e.g. High Dew Point +0.41).

-  $\phi_i < 0$ : Feature reduced the model's predicted risk (e.g. High Barometric Pressure -0.28).

## 2. Precise Viva Language:

*“TreeSHAP explains the model's prediction by identifying which features contributed positively or negatively to the risk score. It does not claim physical causality.”*

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### # CHAPTER 12: WARNING ENGINE & FARMER AI HUB

#### 1. Operational Warning Levels:

- **NO\_WARNING** \to **ADVISORY** \to **WATCH** (Triggered at  $\tau = 0.015$ ) \to **WARNING** (Triggered at  $p \geq 0.05$ ).

#### 2. Farmer AI Knowledge Hub:

- **35,857 verified agronomic records** across 18 JSON files in [src/data/raiKnowledge/](#).

- **1.75x Farmer Persona Boosting** for agricultural queries.

- Understands Hindi/Hinglish agricultural terms (\*sinchai\*, \*jalbharav\*).

- Grounds advice in specific soil types (Clayey vs. Sandy loam) and crop stages (Sowing, Vegetative, Flowering, Harvest).

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### # CHAPTER 13: DATA INGESTION

1. **Open-Meteo REST API**: Real-time hourly numerical weather forecast inputs for 2m temperature, dew point, surface pressure, cloud cover, wind speed, and precipitation.

2. **NASA GPM IMERG 0.1 deg Grid**: Satellite remote-sensing precipitation data.

3. **Offline Fallback Heuristics**: If external APIs are unreachable during a demo, R.A.I. automatically uses seasonal climatology fallbacks so the application never crashes.

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### # CHAPTER 14: DATABASE ARCHITECTURE

Supabase Cloud PostgreSQL hosts 6 relational tables:

1. `users`: UUID, email, salted password hash, role (`user`/`farmer`).

2. `sessions`: 256-bit token hash, expiry timestamp, revocation status.

3. `otp_challenges`: Salted HMAC code hash, attempts remaining (max 5), single-use status (`is_consumed`).

4. `farmer_profiles`: Village, district, soil type, irrigation source.

5. `farm_plots`: Plot name, acreage, active crop, coordinates.

6. `user_preferences`: Language (`en/hi`), default city node.

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### # CHAPTER 15: RUNNING THE APPLICATION LOCALLY

#### Terminal 1 - Start FastAPI Backend:

```
.\.venv\Scripts\Activate.ps1
python -m uvicorn ml.server:app --host 127.0.0.1 --port 8000
```

#### Terminal 2 - Start React Frontend:

```
npm run dev
```

- **Frontend URL**: `http://localhost:3000`

- **Backend Swagger Docs**: `http://127.0.0.1:8000/docs`

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### # CHAPTER 16: "WHY DID WE CHOOSE THIS?"

- **Why XGBoost over Deep Learning?** Tabular weather data performs best on Gradient Boosted Trees, runs efficiently on CPUs without GPUs, and supports exact TreeSHAP math.

- **Why TreeSHAP over LIME?** TreeSHAP provides exact, deterministic mathematical Shapley attributions without random sampling approximations.
- **Why Platt Probability Calibration?** Raw tree margins are distorted under severe class imbalance. Calibration ensures a 10% predicted probability matches a 10% empirical occurrence rate.
- **Why tau = 0.015 instead of 0.50?** Extreme events occur only 0.44% of the time. Setting tau = 0.015 yields 52.1% recall rather than missing every flood.
- **Why Git + GitHub instead of Docker?** Direct local environments provide lightweight, instant execution and transparent debugging for our hackathon prototype.

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#### # CHAPTER 17: RAPID-FIRE VIVA QUESTION BANK

1. **What is the SIH problem statement?** \to SIH1521 (Explainable AI for Heavy Rainfall Prediction).
2. **What is the heavy rain definition?** \to  $\geq 64.5$  mm/day (IMD Standard).
3. **What is the primary ML algorithm?** \to XGBoost with Platt Sigmoid Probability Calibration.
4. **What is the operational decision threshold?** \to tau = 0.0150 (1.5% calibrated probability).
5. **What is the test ROC-AUC?** \to 0.9605 (Measures ranking/discrimination power).
6. **What is the test PR-AUC?** \to 0.1082 (24.6x lift over baseline).
7. **What is the calibrated Brier score?** \to 0.0044 (Rated EXCELLENT reliability).
8. **What is the validation recall at tau=0.015?** \to 52.1% of heavy rainfall events caught.
9. **What Explainable AI method is used?** \to TreeSHAP computing additive feature attributions.
10. **What is the base value phi\_0?** \to -4.852 log-odds (training distribution average).
11. **How many total records were used for training/testing?** \to 175,440 hourly records.
12. **How many features are in the production model?** \to 14 production inference features (27 in registry).
13. **What backend framework is used?** \to Python FastAPI running on Uvicorn (Port 8000).
14. **What frontend stack is used?** \to React 18 + TypeScript + Vite (Port 3000).
15. **What cloud database is used?** \to Supabase Cloud PostgreSQL with 6 relational tables.
16. **How does OTP delivery work?** \to 6-digit code, salted HMAC hash, sent via Brevo email (Sender ID 1).
17. **How many agronomic records are in Farmer AI?** \to 35,857 records across 18 JSON files.
18. **How many automated backend tests exist?** \to 26 pytest test cases (100% passing).
19. **How is version control managed?** \to Git + GitHub.
20. **What makes R.A.I. unique?** \to It transforms black-box predictions into transparent, physics-grounded civil actions.

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#### # CHAPTER 18: TOP 20 MUST-KNOW QUESTIONS FOR THE PRESENTATION

\*(Refer to Chapter 17 for detailed answers to the 20 primary presentation questions)\*.

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#### # CHAPTER 19: 6-MEMBER TEAM ROLE DISTRIBUTION

Member 1: Team Lead & Presentation -> SIH1521 Mission, 5 Pillars, UI Demonstration  
 Member 2: ML & Data Lead -> XGBoost, Calibration, Data Splitting, Metrics  
 Member 3: Explainable AI Lead -> TreeSHAP Mathematics, Feature Attributions  
 Member 4: Frontend Lead -> React 18, 4-Layer Inspector, Doppler Radar  
 Member 5: Backend & API Lead -> FastAPI, Pydantic Models, Weather Ingestion  
 Member 6: Security & Cloud Lead -> Supabase PostgreSQL, Brevo OTP, Git/GitHub

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#### # CHAPTER 20: 1-PAGE MASTER CHEAT SHEET

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| R.A.I. 1-PAGE VIVA MASTER CHEAT SHEET |
+-----+
```

```
| Problem Statement | SIH1521 - Explainable AI for Heavy Rainfall Prediction |
| Heavy Rain Cutoff | >= 64.5 mm / 24 hours (IMD Standard) |
| Historical Freq. | 0.44% of days (Severe Class Imbalance: 1 in 227) |
| ML Algorithm | XGBoost + Platt Sigmoid Probability Calibration |
| Operating Point | tau = 0.015 (1.5% probability) -> 52.1% Flood Recall |
| ROC-AUC / PR-AUC | ROC-AUC: 0.9605 | PR-AUC: 0.1082 (24.6x Baseline Lift) |
| Brier Score | 0.0044 (Rated EXCELLENT Probability Reliability) |
| XAI Engine | TreeSHAP (Additive Shapley Feature Attributions) |
| 4-Layer Inspector | Observed -> Predicted -> Explained -> Operational Action|
| Data Feeds | Open-Meteo NWP + NASA GPM IMERG 0.1 deg Satellite Grid |
| Full Stack Ports | React 18 (Port 3000) <-> Python FastAPI (Port 8000) |
| Cloud Database | Supabase Managed PostgreSQL (6 Relational Tables) |
| OTP Delivery | Brevo Transactional Email REST API (Sender ID: 1) |
| Farmer AI Hub | 35,857 Records, 1.75x Persona Boost, Hinglish Synonyms |
| Version Control | Git + GitHub (Direct Local Environments - NO DOCKER) |
| Verified Tests | 26 / 26 Backend Pytest Tests Passed (100% Pass Rate) |
+-----+
```